HydraR: Stateful Agentic Orchestration for Scientific Reproducibility in R

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Summary

`HydraR` is a lightweight, state-managed framework designed to orchestrate complex "agentic" workflows that are autonomous, multi-step processes driven by Large Language Models (LLMs) natively within R. While mainstream orchestration tools such as LangChain (Chase, 2022) and CrewAI (Moura, 2023) are predominantly Python-based, `HydraR` fulfills the R community's critical requirements for rigorous reproducibility, auditability, and seamless integration with established statistical pipelines.

Researchers can architect workflows as Directed Acyclic Graphs (DAGs) or iterative state machines, where each node represents an LLM-prompted task, a deterministic R function, or an autonomous auditor. `HydraR` provides a provider-agnostic interface that supports cloud APIs (Gemini, Claude, OpenAI, GitHub Copilot) and local deployments via Ollama. Workflow topologies are defined via declarative YAML manifests that embed `Mermaid.js` syntax for visualization and conditional edge logic (e.g., "if fail, go to Node B") for reactive routing. To ensure reliability, a built-in validation engine performs deep static analysis of these manifests—executing syntactic parse checks, and verifying topological synchronization. While `HydraR` permits cyclic dependencies to facilitate agentic self-correction, it enforces runtime safety through a configurable execution "fuse" (limiting transitions to 25 steps by default) and persistent state checkpointing.

Statement of Need

The contemporary R ecosystem lacks a unified framework providing both high-level agentic orchestration and robust, low-level filesystem safety. While existing

packages, such as `ellmer` (Wickham et al., 2025), focus primarily on conversational interfaces and `mall` (Couch, 2024) excels at mapping LLM inferences over structured data, neither is equipped to manage the complex state persistence and isolated execution environments necessitated by long-running, multi-agent collaborations.

`HydraR` addresses three fundamental challenges in this domain:

1. **Auditability**: It implements a centralized `AgentState` with persistent checkpointing (via DuckDB Raasveldt & Mühleisen, 2019), yielding a complete, verifiable audit trail of all LLM interactions and state mutations.

2. **Filesystem Safety**: Leveraging Git worktrees, `HydraR` isolates parallel agent executions into temporary branches, thereby virtually eliminating data corruption and race conditions during concurrent operations.

3. **Portable Logic Topologies**: By defining workflows via `Mermaid.js` syntax and `YAML` manifests, `HydraR` entirely decouples orchestration intent from the underlying execution engine. This declarative schema functions as an "OpenAPI for Agents" — an intermediate, engine-agnostic representation that not only ensures robust auditability in R but also serves as a highly portable compilation target for broader, code-first frameworks such as LangGraph or Google's Agent Development Toolkit.

State of the Field

Although `reticulate` (Ushey et al., 2024) enables R users to interface with Python frameworks like LangChain, it introduces a mismatch between the two languages' data models when serializing complex R objects across language boundaries. `HydraR` avoids this by building natively on the `R6` object-oriented system (Chang, 2021), preserving the fidelity of statistical objects while maintaining direct compatibility with R's integrated debugging toolchains. Furthermore, unlike `ellmer` (Wickham et al., 2025) (which offers chat-centric workflows without DAG orchestration), and `gptstudio` (Nivard & Wade, 2024) (which serves as an IDE-centric coding assistant), `HydraR` is specifically engineered to manage the full life-cycle of autonomous, parallel, multi-agent workflows with robust state retention.

Software Design

`HydraR` employs a highly modular `R6` architecture spanning several key components:

* `AgentLLMNode` / `AgentLogicNode`: Encapsulate LLM-driven prompts and deterministic pure-R logic steps, respectively. They utilize pluggable `AgentDriver` backends for provider-agnostic model communication.

* `AgentDAG`: The core orchestration engine that manages the node dependencies, validates network topology (via `igraph` (Csárdi & Nepusz, 2006)), and executes logic in parallel (via `furrr` (Vaughan & Reznik, 2022)). It features an integrated compiler-like Validation Engine that treats the YAML/Mermaid manifest as a strict Domain Specific Language (DSL). This engine cross-references Mermaid topologies with `YAML` resource definitions and lints embedded R logic for regulatory compliance (e.g., prohibiting imperative loops), effectively positioning `HydraR` as a rigorous reference implementation for executing portable agentic protocols.

* `AgentState`: A centralized state repository employing versioned history and custom reducers to coordinate multi-agent outputs systematically. - `WorktreeManager`: An engine that provisions isolated Git worktrees for parallel execution tasks, paired with an extensible `ConflictResolver` that performs automated semantic resolution alongside human-in-the-loop task reconciliation.

Research Applications

Multimodal Travel Itinerary Planner

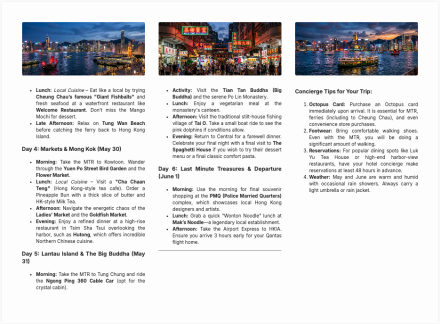
In this example, an LLM `Planner` agent generates a draft itinerary, which is then audited by a deterministic `Validator` against hard constraints. Successfully validated plans proceed to a *multimodal image generation* phase where **Gemini 3.1 Flash** (Google, 2025) creates visual assets based on the itinerary's locales. Finally, a logic node renders a bespoke, CSS-styled HTML pamphlet. Crucially, **conditional looping** between the planner and validator enforces an iterative refinement cycle until all strict parameters are met (Figure 1).

Generated a multi-modal travel pamphlet for a Sydney to Hong Kong itinerary.

Declarative Workflow Excerpt

```
graph: |
  graph LR
    Planner["Travel Planner | type=llm | role_id=travel_concierge"]
    Validator["Constraint Auditor | type=logic | logic_id=validate_constraints"]
    ImageGate["Image Gate | type=logic | logic_id=check_image_status"]
    ImageGenerator["Image Generator | type=logic | logic_id=generate_and_save_images"]
    TemplateManager["Template Provider | type=logic | logic_id=provide_template"]
    PamphletFormatter["Pamphlet Formatter | type=logic | logic_id=format_pamphlet"]
    Finalizer["Itinerary Saver | type=logic | logic_id=save_itinerary"]
```

(a) Travel pamphlet visualization.



(b) Final CSS-styled itinerary.

Figure 1: Multi-modal travel itinerary planner results: (a) generated visual content; (b) final formatted itinerary.

```
    Planner --> Validator
    Validator -- "fail" --> Planner
    Validator -- "pass" --> ImageGate
    ImageGate --> ImageGenerator
    ImageGate --> TemplateManager
    ImageGenerator --> TemplateManager
    TemplateManager --> PamphletFormatter
    PamphletFormatter --> Finalizer

roles:
  travel_concierge: >
    You are a professional travel concierge...
```

Figure 2: Visual DAG structure of the multi-modal travel itinerary planner, featuring iterative validation loops and parallel generation paths.

R Orchestration (Minimal)

```r
library(HydraR)
wf <- load_workflow("hong_kong_travel.yml")
dag <- spawn_dag(wf, auto_node_factory())
results <- dag$run(initial_state = wf$initial_state)
```

Parallel Sorting Algorithm Comparison

This benchmarking example tasks three distinct LLM agents with simultaneously implementing different sorting algorithms. `HydraR` executes each task within an isolated `Git worktree` to aggressively uncouple filesystem side-effects and prevent merge conflicts. Following execution, a Merge Harmonizer node systematically reconciles the independent branches back to the main state, preceding a terminal logic node that empirically benchmarks the aggregated algorithms across five continuous trials (Figure 3).

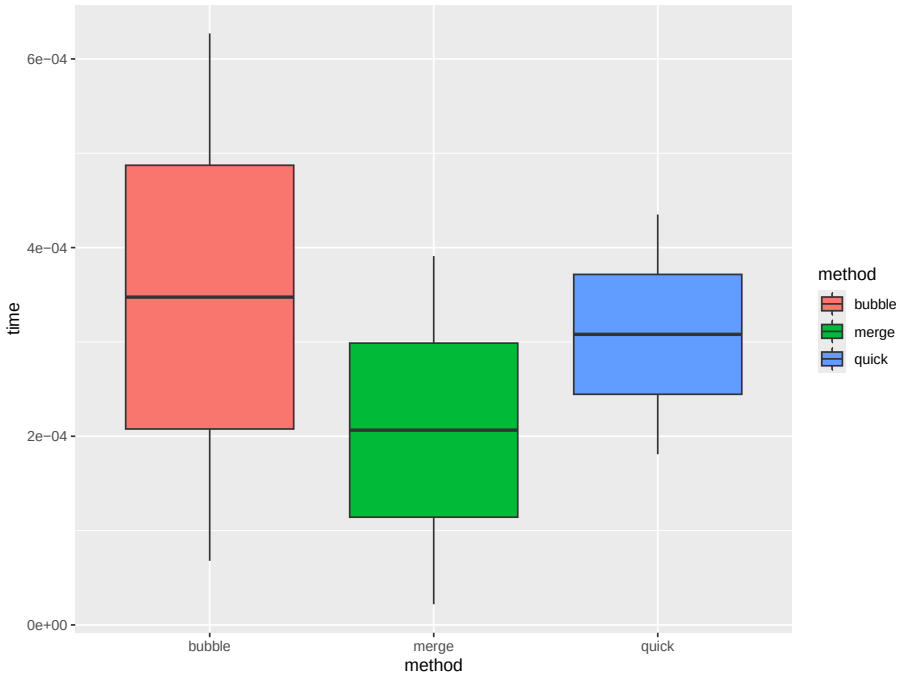


Figure 3: Sorting Algorithm Performance Benchmark (1,000 elements over 5 trials).

Declarative Workflow Excerpt

```
graph: |
  graph LR
      bubble["Bubble Agent | type=llm | role_id=bubble"]
      quick["Quick Agent | type=llm | role_id=quick"]
      merge["Merge Agent | type=llm | role_id=merge"]
      merger["Merge Harmonizer | type=merge"]
      benchmark["Benchmark | type=logic | logic_id=run_benchmark"]
```

```
    bubble --> merger
    quick --> merger
    merge --> merger
    merger --> benchmark
```

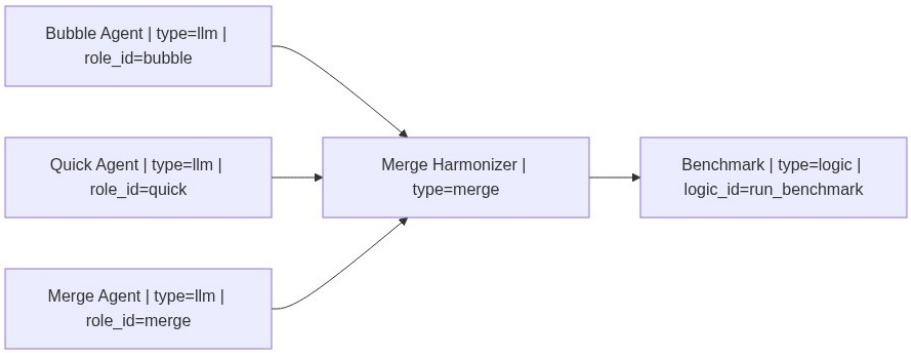


Figure 4: Visual representation of the parallel sorting comparison workflow, showing the fan-out/fan-in pattern and the merge harmonizer.

R Orchestration (Parallel with Isolation)

```r
library(HydraR)
wf <- load_workflow("sorting_benchmark.yml")
dag <- spawn_dag(wf, auto_node_factory())
results <- dag$run(initial_state = wf$initial_state, use_worktrees = TRUE)
```

Fault-Tolerant Pipelines with DuckDB Persistence

To mitigate transient failures inherent to long-running scientific workflows, `HydraR` integrates advanced checkpoint-and-resume functionality powered by DuckDB (Raasveldt & Mühleisen, 2019). Whenever a pipeline pauses mid-execution, it automatically persists its complete `AgentState`. Upon restart, any operator-applied programmatic interventions are seamlessly merged, actively preventing redundant evaluations by allowing execution to jump directly back to the paused node.

Declarative Workflow Definition

```yaml
graph: |
  graph LR
```

```
    Step1["Initialization"] --> Step2["Risky Logic"]
    Step2["Risky Logic"] --> Step3["Conclusion"]
logic:
  check_fixed: |
    if (!isTRUE(state$get("fixed")))
      return(list(status = "PAUSE", output = "Waiting for manual fix."))
    list(status = "SUCCESS", output = "System recovered!")
```

Figure 5: Visual logic of the fault-tolerant pipeline with DuckDB persistence, illustrating a linear flow with restartable checkpoints.

Checkpoint and Resume

```r
library(HydraR)
wf    <- load_workflow("state_persistence.yml")
dag   <- spawn_dag(wf)
saver <- DuckDBSaver$new(db_path = "history.duckdb")
tid   <- "reprex-session-001"

# Run 1: Pauses at Step2 - state saved to DuckDB
res1 <- dag$run(thread_id = tid, checkpointer = saver,
                initial_state = list(fixed = FALSE))

# Run 2: Fix applied, resume from Step2
res2 <- dag$run(thread_id = tid, checkpointer = saver,
                initial_state = list(fixed = TRUE),
                resume_from = "Step2")
```

By safeguarding computational progress autonomously, this pattern effectively circumvents prohibitively costly re-execution in data science pipelines reliant on voluminous LLM invocations or laborious large-scale database queries.

Research Impact Statement

`HydraR` is actively deployed at the **Australian Proteome Analysis Facility (APAF)** to architect and automate resilient bioinformatics tooling. Its declarative YAML schemas empower non-programming domain experts to routinely

govern advanced LLM logic pipelines without any R coding proficiency, thereby democratizing the scale and reproducibility of model-assisted scientific research. Furthermore, by championing a "State-Machine-as-a-Document" standard over proprietary, code-bound execution graphs, `HydraR` pioneers a universal, engine-agnostic blueprint for multi-agent workflows. This standardized format bridges the visualization gap between domain researchers and software engineers, ensuring complex protocols remain infinitely auditable, version-controllable, and portable across disparate AI ecosystems.

AI Usage Disclosure

Transparency and accountability are core tenets of the `HydraR` paradigm. A substantial portion of the implementation was co-developed utilizing **Antigravity**, an agentic AI coding assistant, supervised by a rigorous "human-in-the-loop" review protocol. Specifically, the original authors independently conceptualized the underlying system architecture, while the AI generated specialized logic, unit tests, and software documentation with every resultant commit sequentially tested and manually verified.

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